

Sums, mixtures and normal distributions.

Covered by chapter 3.5-3.6 in the textbook.

Sums and linear combinations

Simulating sums of random variables is straightforward. We have already done this many times (e.g. number of head, sums of dice). See however example 3.10 in the book and the first part of the R code `sums_mixtures_examples.R` for efficient implementation of this in R, avoiding `for`-loops (which are slow in R).

Notice also that a sum $S = X_1 + X_2 + \dots + X_m$ of several iid random variables is sometimes referred to as a *convolution* (since the pdf/pmf for the sum is found by a convolution of the pdfs/pmfs of the individual terms).

Simulating a general linear combination,

$$Y = a_1X_1 + a_2X_2 + \dots + a_mX_m = \sum_{i=1}^m a_iX_i,$$

is also easy as long as the X 's are independent.

Example: Consider the linear combination

$$Y = 10X_1 + 25X_2 + 100X_3$$

where

- $X_1 \sim N(\mu = 20, \sigma^2 = 100)$
- $X_2 \sim \text{gamma}(\alpha = 3, \beta = 2)$
- $X_3 \sim \text{beta}(\alpha = 2, \beta = 2)$

and X_1 , X_2 and X_3 are independent.

Find $E(Y)$, $\text{Var}(Y)$, $P(Y > 200)$, $P(Y < 300)$
and $P(200 < Y < 300)$.

See R code: `sums_mixtures_examples.R`

Mixtures

Mixtures refer to situations where a population consists of several sub populations with different distributions. Each observation come from one of the sub populations.

The pdf/pmf of the total distribution can be written

$$f_X(x) = p_1 f_{X_1}(x) + p_2 f_{X_2}(x) + \dots + p_m f_{X_m}(x)$$

where p_i is the probability of coming from distribution i and $f_{X_i}(x)$ is the pdf/pmf of distribution i .

Example: The height of Norwegian women $X_1 \sim N(168, 6.0^2)$ and men $X_2 \sim N(180, 6.5^2)$. I.e. the height of adult Norwegians is a mixture of two normal distributions.

See R code: `sums_mixtures_examples.R`

Normal distribution

The normal distribution is the most important distributions, and neither the inverse transform nor the acceptance-rejection method works efficiently for simulating from this distribution.

Univariate normal distribution

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}, \quad -\infty < x < \infty,$$

For the ordinary $N(\mu, \sigma^2)$ -distribution first notice that it is sufficient to be able to simulate from the $N(0, 1)$ -distribution since if $Z \sim N(0, 1)$ then $X = \mu + \sigma Z \sim N(\mu, \sigma^2)$.

One algorithm for the $N(0, 1)$: Box-Muller algorithm to produce independent $x, y \sim N(0, 1)$.

1. Generate $\theta \sim U[0, 2\pi]$.
2. Generate $u \sim U[0, 1]$ and let $r = \sqrt{-2 \ln(1 - u)}$.
3. Let $x = r \cos(\theta)$ and $y = r \sin(\theta)$. □

However, do not implement Box-Muller or other normal random number algorithms directly in R. Use `rnorm(n)` instead, as this relies on highly optimized algorithms/implementations.

Multivariate normal distribution

Example: In economy models it is very common to work with dependent normally distributed random variables. E.g. consider three investments where the surplus is modelled:

- $X_1 \sim N(\mu_1 = 8, \sigma_1^2 = 100)$
- $X_2 \sim N(\mu_2 = 4, \sigma_2^2 = 16)$
- $X_3 \sim N(\mu_3 = 1.05, \sigma_3^2 = 0.25)$
- $\rho_{12} = \rho(X_1, X_2) = -0.6$
- $\rho_{13} = \rho(X_1, X_3) = 0.3$
- $\rho_{23} = \rho(X_2, X_3) = -0.4$

We might be interested in various probabilities, or considerations around the surplus of investment portfolios like

$$Y = aX_1 + bX_2 + cX_3$$

for different values of a , b and c .

This can be done by calculations or by simulating from the specified multivariate normal distribution.

The multivariate normal distribution was briefly described in chapter 2. Generally for $\mathbf{X} = (X_1, X_2, \dots, X_d)^T$ the pdf is

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\mathbf{\Sigma}|^{1/2}} \exp\{-0.5(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})\}$$

Where:

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_d \end{bmatrix}, \quad \mathbf{\Sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1d} \\ \sigma_{21} & \sigma_{22} & \dots & \sigma_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{d1} & \sigma_{d2} & \dots & \sigma_{dd} \end{bmatrix}$$

E.g. $\mu_i = E(X_i)$, $\sigma_{ii} = Var(X_i)$, $1 \leq i \leq d$
 $\sigma_{ij} = Cov(X_i, X_j) = Cov(X_j, X_i)$, $1 \leq i, j, \leq d$
 Notation often used: $\mathbf{\Sigma} = Cov(\mathbf{X})$

We write $\mathbf{X} \sim N_d(\boldsymbol{\mu}, \mathbf{\Sigma})$.

The multivariate normal has lots of applications, e.g. in economy, physics, engineering, signal processing, pattern recognition, etc.

How to simulate from the multivariate normal?

We first define the standard multivariate normal distribution where $\mathbf{Z} = (Z_1, Z_2, \dots, Z_d)^T$ is a vector of iid standard normal distributed variables with

$$\boldsymbol{\mu}_{\mathbf{Z}} = \begin{bmatrix} 0 \\ 0 \\ \vdots \end{bmatrix}, \quad \boldsymbol{\Sigma}_{\mathbf{Z}} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

To simulate from the standard multivariate normal we just simulate d iid observations from the univariate standard normal distribution.

Further, similar as for the univariate case it can be show that for a linear transformation:

$$\mathbf{C}\mathbf{Z} + \mathbf{b} \sim \text{N}_d(\mathbf{b}, \mathbf{C}\mathbf{C}^T)$$

Then by letting $\mathbf{b} = \boldsymbol{\mu}$ and if we can find a $\mathbf{C}$ such that $\mathbf{C}\mathbf{C}^T = \boldsymbol{\Sigma}$ we achieve

$$\mathbf{C}\mathbf{Z} + \boldsymbol{\mu} \sim \text{N}_d(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

There are several techniques in linear algebra for decomposing a matrix Σ such that we get $\mathbf{C}\mathbf{C}^T = \Sigma$ (spectral decomposition, singular value decomposition or Choleski decomposition) - see chapter 3.6.1 in the book. Using one of these decompositions the algorithm becomes:

1. Generate a vector $\mathbf{z} = (z_1, z_2, \dots, z_d)^T$ of d independent standard normal variables.
2. Return $\mathbf{x} = \mathbf{C}\mathbf{z} + \boldsymbol{\mu}$. □

For simulating n variables from the $N_d(\boldsymbol{\mu}, \Sigma)$ distribution, instead of repeating the algorithm n times in a loop, the steps above can be extended by matrix/vector operations to do all simulations in one go. See page 70-71 in the book.

In R the package `mvtnorm` include functions for simulating from the multivariate normal distribution. All the three decompositions mentioned above are implemented, see the help pages for `rmvnorm`.

Example: Recall the economy example. We will now use simulations to:

- Estimate probabilities, e.g. the probability that all of the investments have a positive surplus.
- Evaluating various investments strategies yielding a surplus $Y = aX_1 + bX_2 + cX_3$

Investing in 1 costs 10 per unit, 2 costs 5 per unit and 3 costs 1 per unit, and in total we can invest 300. I.e. we have the restriction that $10a + 5b + c \leq 300$.

For the simulation we need to specify $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. From page 6, $\boldsymbol{\mu} = (8, 4, 1.05)^T$.

To find $\boldsymbol{\Sigma}$ recall that

$$\rho_{12} = \text{Corr}(X_1, X_2) = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Var}(X_1)\text{Var}(X_2)}} = \frac{\sigma_{12}}{\sigma_1\sigma_2}$$

which implies

$$\sigma_{12} = \rho_{12}\sigma_1\sigma_2$$

Thus, using the information on page 6:

$$\sigma_{11} = \sigma_1^2 = 100$$

$$\sigma_{22} = \sigma_2^2 = 16$$

$$\sigma_{33} = \sigma_3^2 = 0.25$$

$$\sigma_{12} = \sigma_{21} = -0.6 \cdot 10 \cdot 4 = -24$$

$$\sigma_{13} = \sigma_{31} = 0.3 \cdot 10 \cdot 0.5 = 1.5$$

$$\sigma_{23} = \sigma_{32} = -0.4 \cdot 4 \cdot 0.5 = -0.8$$

I.e.

$$\mathbf{\Sigma} = \begin{bmatrix} 100 & -24 & 1.5 \\ -24 & 16 & -0.8 \\ 1.5 & -0.8 & 0.25 \end{bmatrix}$$

See R code: `multivariate_normal_example.R`

Directional example:

Uniform distribution on a sphere

Let

$$\mathbf{Z} = (Z_1, \dots, Z_d)^T, \quad Z_i \stackrel{\text{iid}}{\sim} N(0, 1).$$

Every non-zero vector can be decomposed into a length and a direction:

$$\mathbf{Z} = RU, \quad R = \|\mathbf{Z}\|, \quad U = \frac{\mathbf{Z}}{\|\mathbf{Z}\|}.$$

For $\mathbf{Z} \sim N_d(\mathbf{0}, I)$, the density depends only on $\|\mathbf{Z}\|$ and not on the direction of $\mathbf{Z}$. Thus the distribution is rotationally symmetric and all directions are equally likely. Consequently,

$$U = \frac{\mathbf{Z}}{\|\mathbf{Z}\|}$$

is uniformly distributed on S^{d-1} .

Normalization removes the random length R and retains only the uniformly distributed direction.

See R code: `SphereDist2.R`